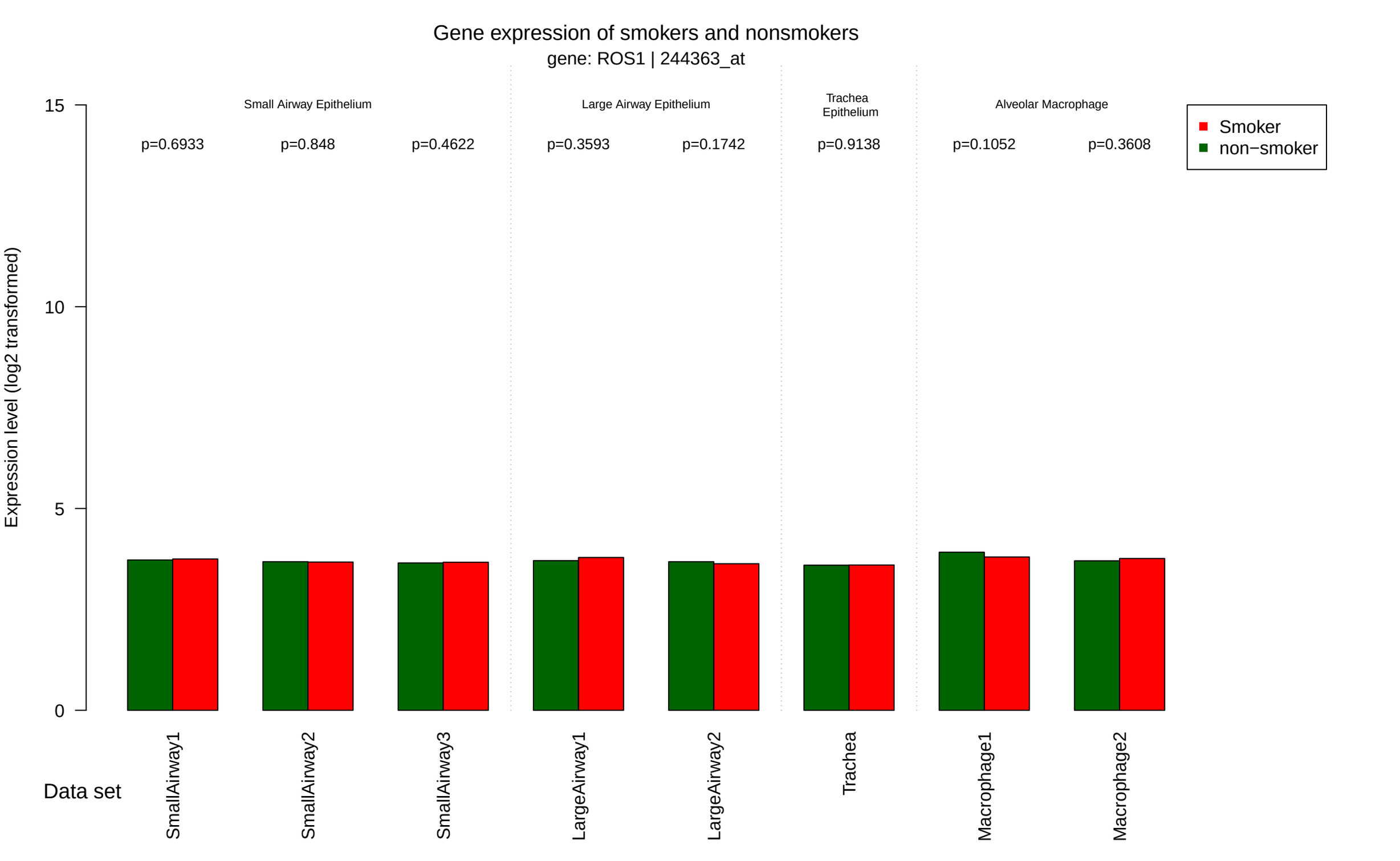
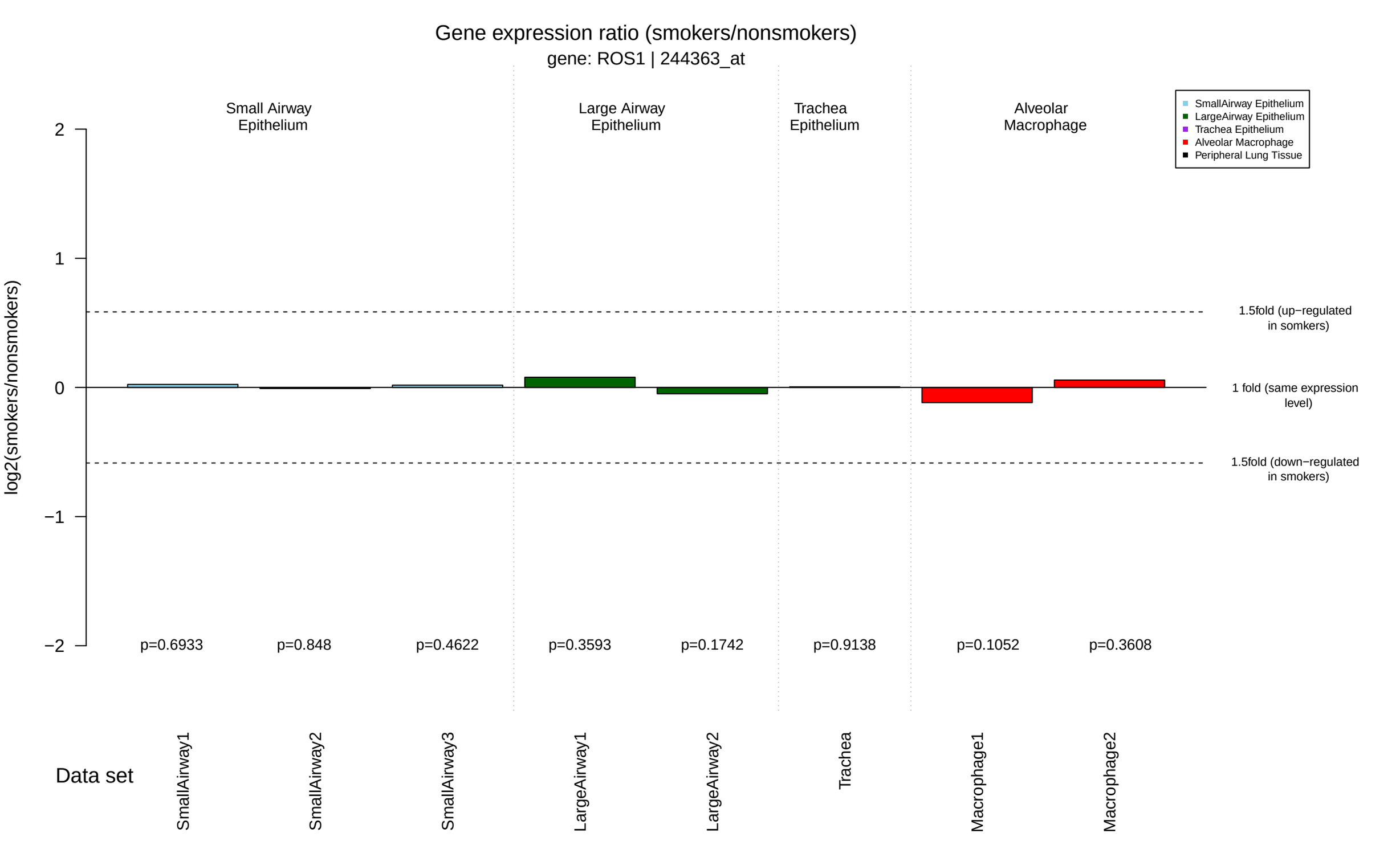
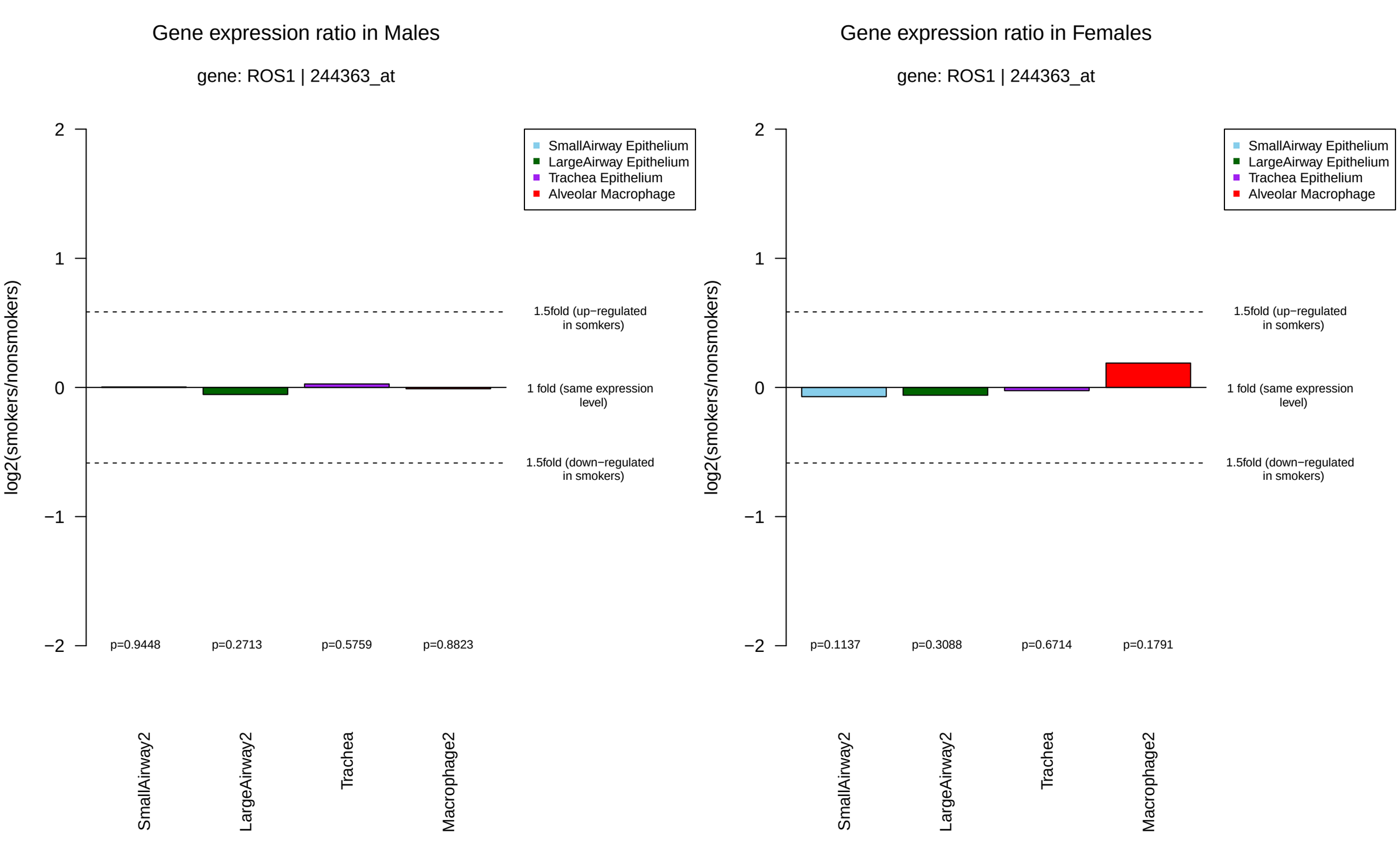
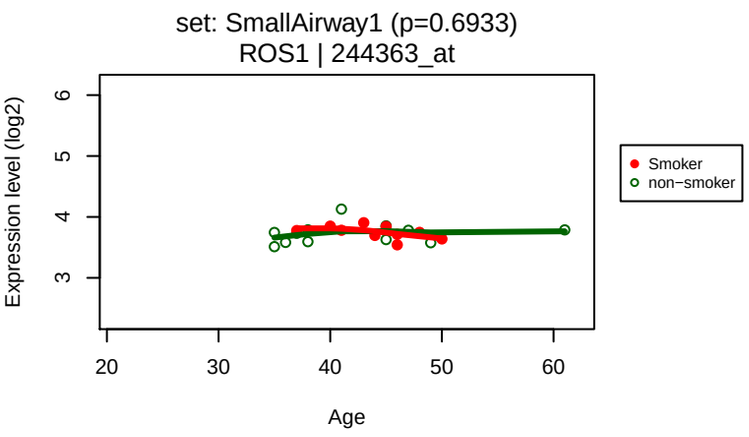
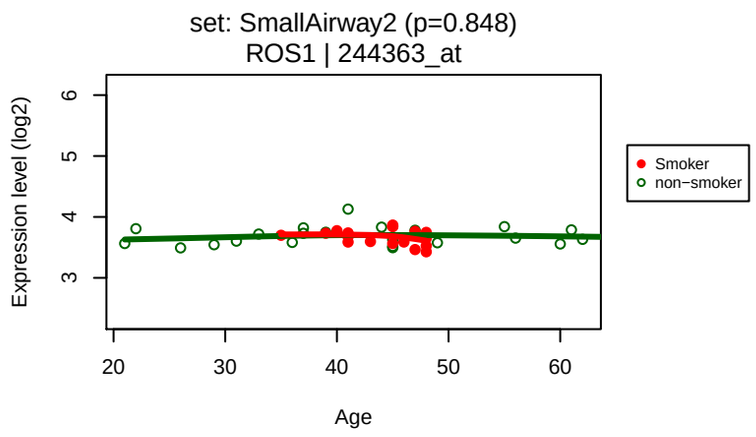
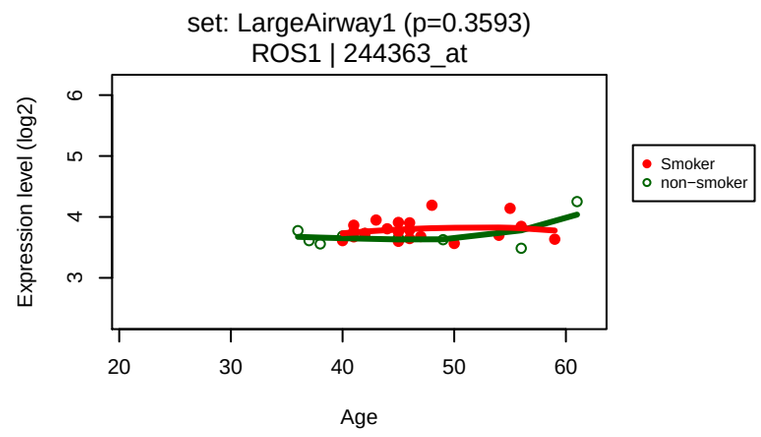

Gene expression of smokers and nonsmokers
gene: ROS1 | 244363_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6933
p=0.848
p=0.4622
p=0.3593
p=0.1742
p=0.9138
p=0.1052
p=0.3608
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

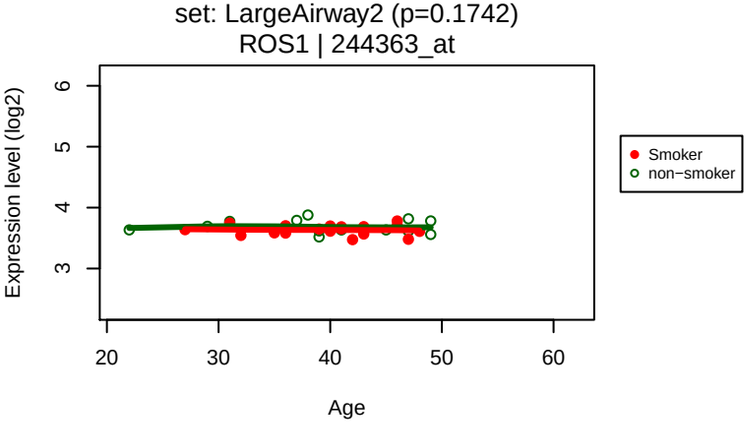

Gene expression ratio (smokers/nonsmokers)
gene: ROS1 | 244363_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6933
p=0.848
p=0.4622
p=0.3593
p=0.1742
p=0.9138
p=0.1052
p=0.3608
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

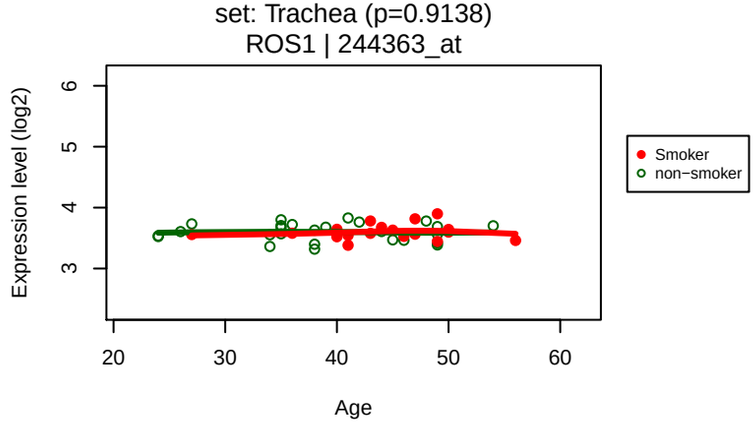

Gene expression ratio in Males
gene: ROS1 | 244363_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9448
p=0.2713
p=0.5759
p=0.8823
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROS1 | 244363_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1137
p=0.3088
p=0.6714
p=0.1791
SmallAirway2
LargeAirway2
Trachea
Macrophage2

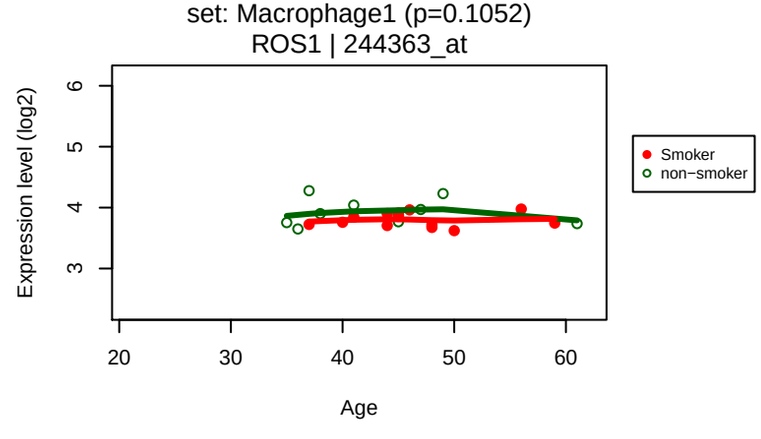

set: SmallAirway1 (p=0.6933)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker

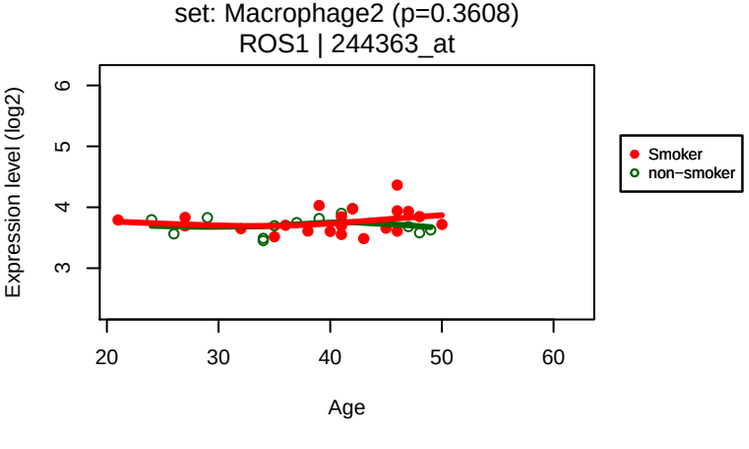

set: SmallAirway2 (p=0.848)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker

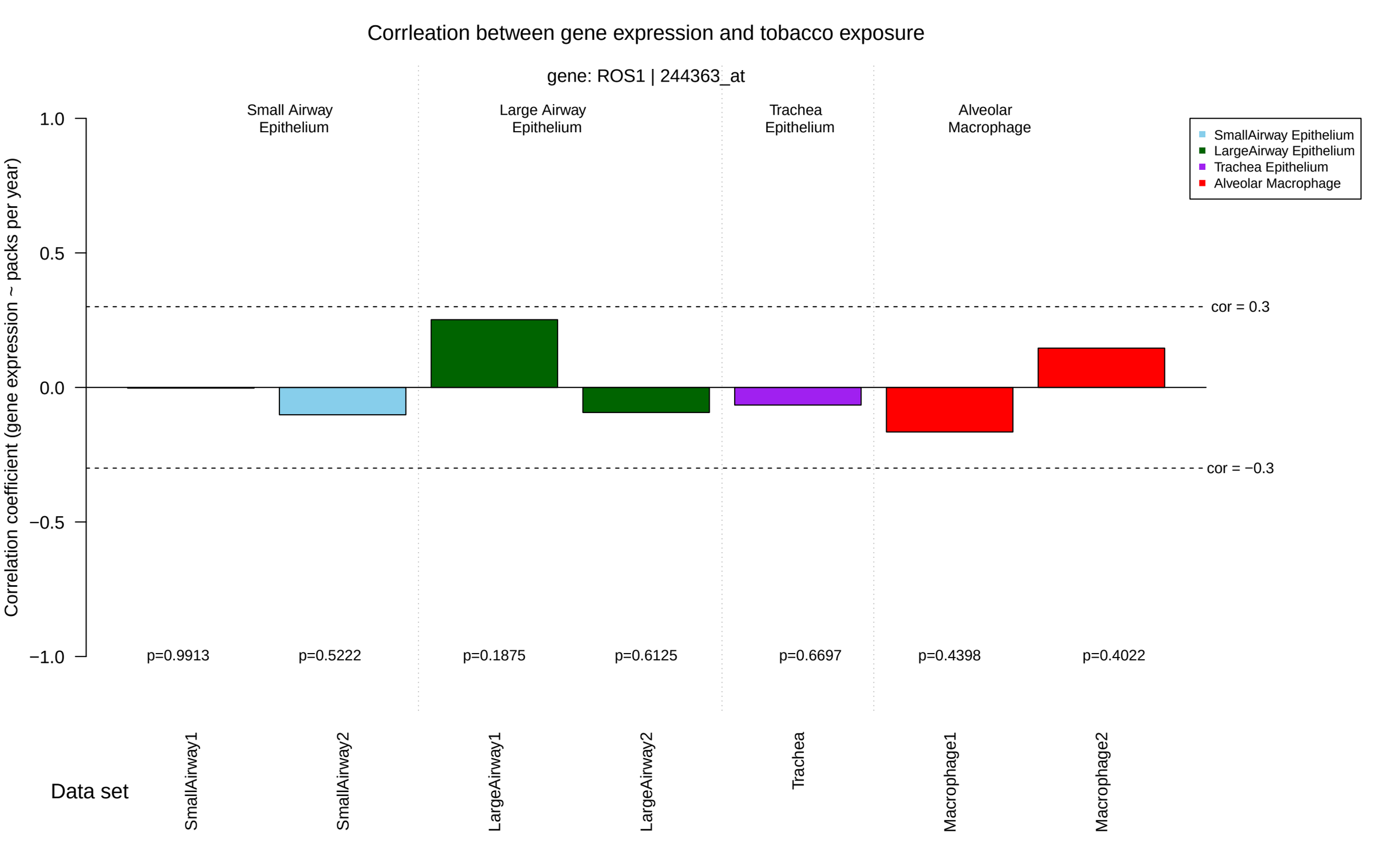

set: LargeAirway1 (p=0.3593)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1742)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9138)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1052)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3608)
ROS1 | 244363_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROS1 | 244363_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9913
p=0.5222
p=0.1875
p=0.6125
p=0.6697
p=0.4398
p=0.4022
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2